

River Phosphate Aspects of Poultry Farming in Powys – Part 2

Following on from my earlier article, I have looked in more depth at approved planning applications for poultry units in Powys over the period 2000 - 2019. By doing this, I have been able to determine:

- *the total number of projects involved*
- *the number and type of birds by project*
- *the timing and location of each project*

From this, I have been able to calculate the phosphate output for each project.

I have then been able to place each project within a river catchment to determine the potential phosphate run-off to each river.

By making use of the time dependency of phosphate emissions within specific river catchments, I have been able to estimate the time dependency of the resulting phosphate run-off rates.

The results clearly demonstrate that phosphate emissions from poultry operations are of exactly the right magnitude to explain the increasing impact of phosphates in our rivers. It is clear that there is only one way out of this mess.

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Introduction

In the previous article I studied a recent Planning Application (PA) for a new broiler poultry unit near Knighton on the river Teme. By making use of the Manure Management Plan (MMP) within the PA, I was able to demonstrate that the phosphate emissions from this one project would constitute a significant contribution to pollution of the river Teme. By extrapolating to the claimed 8.5 million birds in Powys, I then showed that the poultry industry is a significant pollution source for the entire Wye and Severn catchments.

In the first article, I introduced a key conservation concept. All the phosphate spread onto the land is ultimately carried away in surface water (or groundwater, where local geology permits). Storage of phosphate within the soil only affects the transient response of the system. If the phosphate application increases, the soil absorbs additional phosphate only until it comes into equilibrium with the new application rate. It might take a few years to reach equilibrium, but when it does, the phosphate going in must equal the phosphate coming out. There are only two output streams: run-off to the rivers and phosphate exported within the crop. For livestock businesses, the latter is relatively low.

A weakness of the first calculation was scaling up from a broiler unit (broilers are taken as day old chicks and are sold at 6 weeks). It turns out that many poultry units in Powys are for egg production, obviously with more mature birds; this unsurprisingly has an impact on the amount of N and P emitted. There are also breeder units and pullet rearing units, all with different emission figures. In this study I have made use of applicable N emission rates for each project, resulting in more robust P emission numbers.

An unanswered question in the first study was how much phosphate goes into each river. In this work, I have located each project and determined the river which it drains into. Of particular interest is the question regarding Wye algal blooms and their origin on the river Ithon. I have also been able to analyse the temporal development of poultry numbers and thus estimate the resulting development of phosphate concentration.

Geography

Powys does not possess high-grade soil; the vast bulk of the land being used for grazing. The majority of farms associated with poultry projects keep sheep and/or cattle. There is relatively little dairy activity and virtually no arable farming. It is easy to see why poultry is a logical add-on for these businesses.

The geology in the area is such that all rainwater, allowing for evaporation and transpiration losses, drains to the rivers; as far as I can tell, there are no major underground aquifers. By the same token, we can thus assume that any phosphate spread on the land ultimately goes into the rivers. An inspection of the OS mapping for the area shows that Powys is a mass of valleys, each with a river or stream at the bottom. There can hardly be a farm in Powys without a stream running through it.

Most of Powys is drained by the Severn catchment to the north and the Wye catchment south of this. Around the edges we also have the Dee in the extreme north, the Usk in the extreme south and a few westerly flowing rivers such as the Dovey on the western edge. None of these latter rivers will feature significantly in what follows.

The data

Overview

Searching for Powys CC planning applications for all poultry related applications is a slow and laborious job. The only option is to search the *Title* field for likely key words. However, part way through this research, I was introduced to the [Campaign for the Protection of Rural Wales](#) (CPRW), who have been monitoring the Powys poultry industry for many years. Unlike Powys CC and NRW, they do maintain a comprehensive database of relevant planning applications. Their data underpins the analysis which follows – and I am very grateful for being given access to it.

In total, I have been able to include a total of 290 approved projects spanning the period 2000 – 2019. The total number of birds involved is 8.8 million. Data before 2005 is a little sketchy, but from 2005 onwards it is pretty solid.

Planning applications submitted since 2015 have all documents, including MMPs, available online. For about half of these (69 in total) I have downloaded the MMP in order to (a) understand the size and nature of existing operations, and (b) to check the details of the nutrient calculations.

Farm activities

This table shows the breakdown of other farm activities described in the downloaded MMPs.

<i>Non-poultry activity</i>	<i>Number</i>
Cattle (inc. dairy) and/or sheep	59
None (i.e. poultry only)	7
Not declared	3
Total	69

Table 1: farm activities

Very little arable activity was declared beyond silage and other feed crops. As per the case study in my first report [3], the new poultry operation is always additional to existing sheep & cattle activities.

The size of farms varies enormously from less than 50ha up to 500ha; the average though is 150ha.

Disposal of poultry waste

The size of the farm is a key driver in determining whether there is a need to export some of the poultry manure. The larger farms generally have no need to export and can spread without exceeding the 250kgN/ha limit. Of the 69 MMPs examined, 63 are spreading the poultry manure on their own land. Six are ‘exporting’ some or all, but almost certainly to other farms in the area. None of the MMPs examined had any convincing evidence of any poultry manure not being spread directly on land, for example going to Anaerobic Digestion (AD) plants. A very few said that an AD plant would take excess manure, but this was never quantified or committed to.

All the 69 MMPs examined contained calculations to prove compliance with the 250kgN/ha limit. NRW approved every one of them on this basis alone. None of them considered phosphate loading or even acknowledged where the excess phosphate would go.

To underline the cynicism involved, there is a standard calculation often used in MMPs. This is to calculate the area of land required to support the stated number of livestock at 250kgN/ha. If the farm is bigger than that, then fine. The need to balance nutrients to crop requirements, as required by *RB209* [1] and *CoGAP* [2], is simply not addressed in any of the MMPs and clearly is not enforced within the planning process by NRW or by Powys CC.

Farmers are taking advantage of the 250kgN/ha limit, as if it's akin to a business tax allowance.

An aspect worth clarifying at this point regards silage. Most farms in Powys will produce some silage, which has a higher nutrient requirement than pasture. The first point to make is that silage is grown on only a small part of the farm – depending on farm size and stock density, but maybe 20%. The second point is that this silage is not exported: it feeds the stock in winter. This means that the phosphate in the silage is recycled each year: soil to silage to cattle to manure and back to soil. It is not an output stream unless it is exported from the farm. MMPs frequently cite silage as a justification for additional fertiliser, but spookily there are never any numbers to back this up. It is all smoke and mirrors.

Project types

Poultry operations fall into one of four types. *Breeders* produce chicks, generally sold as day-old chicks. *Broiler rearers* take these day-old chicks and fatten them for 6 weeks, selling them for meat at around 2.2kg weight; this process typically repeats 8 times per year. *Pullet rearers* take day-old chicks and grow them for 16 weeks, at which point they are able to lay eggs; this process typically repeats 3 times per year. *Egg producers* take 16-week-old pullets and produce eggs until yields fall after 12-18 months. Almost all egg producers in Powys are free-range, including a handful of organic free-range.

All of the above project types have the birds housed in large industrial sheds. Free-range units merely have *pop-holes* in the walls to allow birds daytime access to a field. All these activities are classified as Intensive Poultry Units (IPUs).

Project sizes vary by type. The majority of egg production units are in multiples of 16,000 birds (organic units are smaller), with most such projects in Powys being for 16,000 or 32,000 birds. Breeders and pullet rearers are generally in the 20,000 to 60,000 range. Broiler units can be huge, with projects in excess of 100,000 birds not uncommon. In all types, multiple projects are often executed cumulatively over time on the same farm.

The breakdown within the full dataset is as follows:

<i>Project type</i>	<i>Number of projects</i>	<i>Number of birds</i>
Breeders	9	220,500
Broiler rearers	63	4,483,700
Egg producer	195	3,242,800
Pullet rearers	23	874,000
<i>Totals</i>	<i>290</i>	<i>8,822,000</i>

Table 2: project types in the full dataset

That's fairly straightforward.

Phosphate analysis

Calculation of emissions

Nitrogen emissions from poultry are well documented and can be found for example in *Table 3* of the *NVZ guidance for farmers* [3]. These are the numbers which underpin all of the MMPs examined. From this value for N, we can use *Tables 2.5 & 2.7* from *RB209* [1] to derive the corresponding P₂O₅ value. These numbers are shown in the following table:

<i>Project type</i>	<i>kgN/year per 1000 birds</i>	<i>kgP₂O₅/year per 1000 birds</i>
Breeders	700	425
Broiler rearers	330	200
Egg producer	530	322
Pullet rearers	210	128

Table 3: N & P emissions from different types of poultry

Knowing the number and type of birds for each planning application, it is then a simple matter to apply the appropriate numbers to calculate the relevant N and P₂O₅ emissions. Within the 69 downloaded MMPs, I've also verified consistency with submitted numbers.

Free range aspects

Our basis for calculation has been of birds housed in IPUs, with all of the manure spread on the land by the farmer. Since a significant number of birds are free range layers, it is worth considering what happens while the birds are outside depositing manure and urine directly on the range area. In the UK, free range means that birds have access to the outside during daylight hours, at a stock density of 2,000 birds per hectare. If we apply this stock density, with the 'egg producer' numbers from Table 3, for 50% of the time, we get manure application rates of 530kgN/ha and 322kgP₂O₅/ha. This is over twice the permitted application rate for organic manure.

Clearly, we don't know what proportion of birds choose to avail themselves of fresh air each day, so these numbers may be a slight overestimate. But it does show that such a range area represents a phosphate hot spot. If this happened to be located too close to surface water, it could cause some significant point pollution. But from the farm boundary point of view, there is no change. The total amount of nutrients applied to the land of any one farm is the same, irrespective of whether the hens deposit it, or the farmer spreads it. On this basis, I will not consider free range to be a special case.

Project locations

It is a straightforward process to take the postcode of each project and locate it on a map. It is then possible to track down the local streams and determine which river the farm drains into. I have categorised this firstly by major river catchment (I've called this *Catchment*; to NRW it is a *Management Catchment*) and secondly by significant tributaries (which I've called *River*; to NRW an *Operational Catchment*).

This is how things break down by major catchment:

<i>Catchment</i>	<i>Number of projects</i>	<i>Number of birds</i>	<i>kgP₂O₅/year</i>
Dee	1	-	-
Dovey	1	36,600	4,658
Severn	146	5,281,000	1,263,413
Wye	142	3,504,400	875,399
<i>Totals</i>	<i>290</i>	<i>8,822,000</i>	<i>2,143,470</i>

Table 4: allocation of projects, birds and phosphate emissions by major catchment

Within the dataset, there are no poultry projects draining into the Usk catchment. The Dee project is not operational and the single Dovey project is small. Therefore, for the remainder of this document I will discuss the 288 projects spread over the Severn and Wye catchments.

Incidentally, it is worth pointing out that my estimate of 850 tonnes p.a. of P₂O₅ in the Wye catchment, as used in my first report [3], was not too far off the mark.

Wye catchment

The next table shows how the Wye catchment projects break down by river:

<i>River</i>	<i>Number of projects</i>	<i>Number of birds</i>	<i>kgP₂O₅/year</i>
Arrow	6	107,500	36,952
Bachawy	1	16,000	5,139
Duhonw	1	100,000	20,000
Edw	4	60,000	19,273
Irfon	3	86,600	13,349
Ithon	96	2,078,700	546,595
Lugg	8	451,600	94,446
Marteg	3	28,000	8,994
Wye	20	576,000	130,652
<i>Totals</i>	<i>142</i>	<i>3,504,400</i>	<i>875,399</i>

Table 5: allocation of projects, birds & phosphate emissions by river for the Wye catchment

It is immediately apparent how the Ithon alone accounts for two thirds of the Wye catchment projects and two thirds of the Wye phosphate emissions. To emphasise the point: that is over 546 tonnes of P₂O₅ being spread within the Ithon catchment every year. Geographically these IPU's are located throughout the valley.

The IPU's allocated to the Wye on the other hand, are predominantly clustered around three areas: Rhayader, Builth Wells and Glasbury.

Severn catchment

Table 5 shows how the Severn catchment projects break down by river:

<i>River</i>	<i>Number of projects</i>	<i>Number of birds</i>	<i>kgP₂O₅/year</i>
Banwy	13	565,600	132,325
Bechan	4	84,000	20,000
Caebitra	2	128,000	22,836
Cain	1	38,000	4,836
Camlad	1	90,000	18,000
Miwl	5	293,800	65,402
Rhyw	8	199,000	47,291
Severn	54	1,290,500	381,982
Teme	19	959,500	193,694
Trannon	1	32,000	10,279
Vyrnwy	38	1,600,600	366,767
<i>Totals</i>	<i>146</i>	<i>5,281,000</i>	<i>1,263,413</i>

Table 6: allocation of projects, birds and phosphate emissions by river for the Severn catchment

With the exception of the river Teme, all of these smaller rivers lie within the area known as the Severn Uplands and combine with the main stem of the Severn upstream of Shrewsbury. The Vyrnwy and Upper Severn together account for around two thirds of the projects.

The river Teme is notable in that all of the projects are clustered in the vicinity of Knighton.

Phosphate concentrations

What remains is to convert the agricultural metric $\text{kgP}_2\text{O}_5/\text{year}$ to the scientific metric mgP/s . First, we multiply by 44% to remove the mass of the five oxygen atoms, then we convert from years to seconds. Finally, we can divide by the flowrate of the respective river to determine its total phosphate concentration¹. River flowrates are annual mean flows taken from the UK [National River Flow Archive](#). These are dependent upon the presence of EA flow measuring stations, which are not always present on smaller streams – hence the blanks in the table below. For the upper Teme I have used my estimate from the first report.

<i>Catchment</i>	<i>River</i>	<i>Phosphate flow (mgP/s)</i>	<i>River flow (m³/s)</i>	<i>Phosphate concentration (mgP/litre)</i>
Wye	Arrow	512	2.4	0.22
	Bachawy	71		
	Duhonw	277		
	Edw	267		
	Irfon	185	10.2	0.02
	Ithon	7,568	8.1	0.93
	Lugg	1,308	5.9	0.22
	Marteg	125		
	Wye	1,809	37.0	0.28
Severn	Banwy	1,893		
	Bechan	277		
	Caebita	316		
	Cain	67		
	Camlad	249		
	Miwl	906		
	Rhyw	655		
	Severn	5,289	14.7	0.36
	Teme	2,682	7.5	0.36
	Trannon	142		
	Vyrnwy	5078	21.5	0.24

Table 7: calculated total phosphate concentrations for rivers where flowrate is known

The difficulty with this data is that it takes no account of confluences, for example where the phosphate load from the Arrow contributes to the total load in the lower Lugg. It is more informative to view this data graphically as follows.

¹ Within this report, phosphate concentration always refers to *total phosphate*. A discussion of the complexities of phosphate transport in rivers and phosphate measurement is beyond the scope of this document. It suffices to point out that EA & NRW measure *active phosphate*, which will always be less or equal to *total phosphate*.

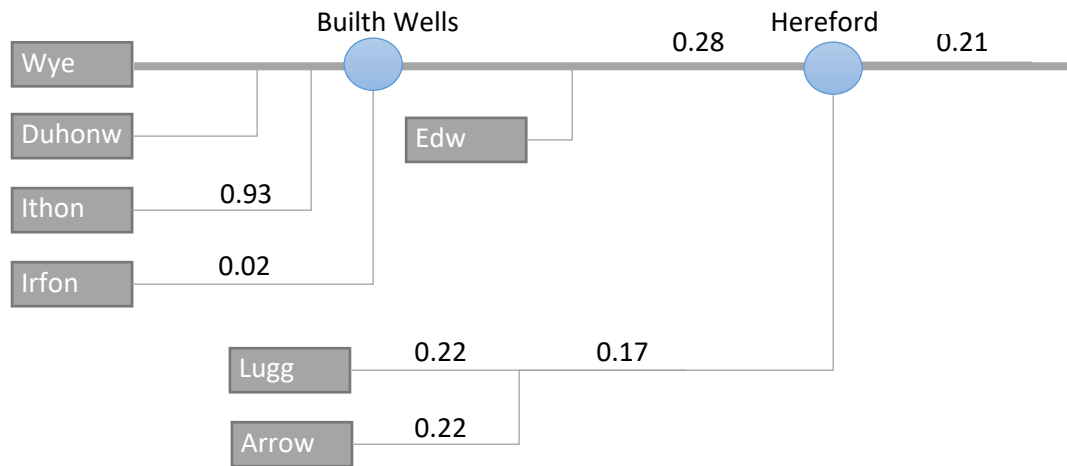


Figure 1: schematic flow-map of Wye catchment showing phosphate loads as mgP/litre

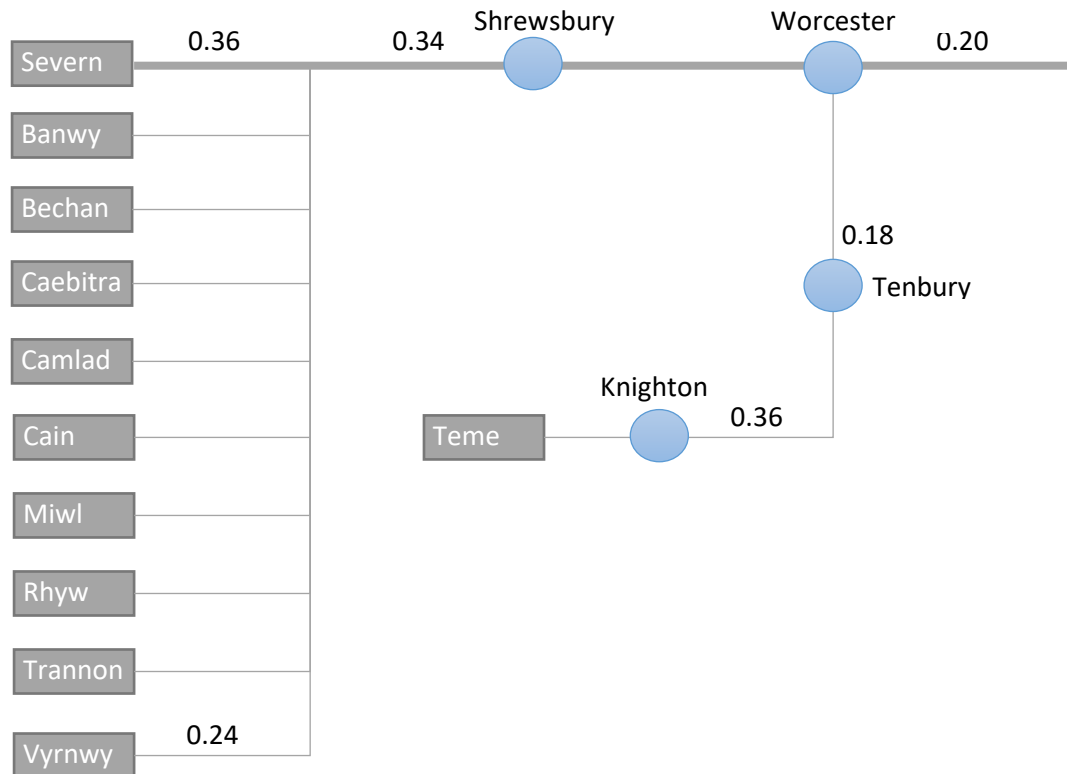


Figure 2: schematic flow-map of Severn catchment showing total phosphate loads as mgP/litre

What these figures show is the predicted total phosphate concentration at different points on the rivers taking account of phosphate load flowing in from their respective tributaries. Where there are no concentrations shown, this is generally because there is no flowrate data available to perform the calculation with.

To interpret these numbers, we first need to make use of the *Water Framework Directive (Standards and Classification) Directions (England and Wales) 2015*; this is the classification system used by EA

and NRW. There are some river-dependent subtleties to this, but the classification is approximately as per the following table:

<i>Class</i>	<i>Min</i>	<i>Max</i>	<i>Description</i>
High	-	0.01	Near natural condition
Good	0.01	0.03	Slight change due to human activities; no impact on fisheries and wildlife
Moderate	0.03	0.09	Moderate change due to human activity; some impact on fisheries and wildlife
Poor	0.09	0.75	Major changes from natural condition; moderate impact on fisheries and wildlife
Bad	0.75	-	Severe changes from natural condition; major impact on fisheries and wildlife; many species absent

Table 11: WFD phosphate classification; units are mgP/litre of reactive phosphate

The stated national target for rivers is *good* – which requires phosphate to be below 0.03mgP/litre, on an annual average basis. Sadly, with the exception of the Irton, all the rivers for which I have been able to show concentrations in figures 1 & 2 fall within the *poor* classification. The Ithon is particularly badly afflicted with a calculated concentration of 0.93 mgP/litre. Within the Severn catchment, the Upper Severn is also notable with a concentration of 0.36 mgP/litre.

Finally, I would emphasise two points.

Firstly, the phosphate concentrations calculated above are purely the additional phosphate resulting from poultry operations. They do not include the baseline phosphate loading from the ‘business as usual’ cattle and sheep rearing activities prevalent in these areas.

Secondly, they are equilibrium values, not necessarily current values. They represent where things are heading if we do nothing. It is important to understand that there is a time element to all of this – a topic which I will explore in the final section.

Here is a summary of the key points from this section:

- *From 2005 up to 2019, around 8.8 million additional birds have been housed in Powys.*
- *The Wye catchment accommodates 3.5 million birds, which emit 875 tonnes of P₂O₅ equivalent per year.*
 - *Of this, the Ithon accommodates 2.0 million birds, which emit 547 tonnes of P₂O₅ equivalent per year; this is a truly huge number on such a small river.*
- *The Severn catchment accommodates 5.3 million birds, which emit 1,263 tonnes of P₂O₅ equivalent per year.*
 - *The upper Severn and Vyrnwy together account for 749 tonnes of P₂O₅.*
- *This amount of phosphate is sufficient to take the majority of affected rivers into Water Framework Directive **poor** classification.*

Time dependency of river phosphate level

In my first report [3], I discussed how soil stores phosphate. The concept of phosphate storage in soil is well accepted and is described, for example, in *RB209* [1]. The time required to fill and empty this storage varies with local conditions but is long – maybe ten years or more. In Appendix 1 of my first report, I explained how we can model this situation mathematically. Here I will do just that, in order to understand how much of the phosphate that has already been spread on the land is getting into the rivers today.

Essentially, the phosphate ‘emissions’ which I calculated in the previous section represent the current amount of phosphate being spread on the land. What currently goes into the river is different because it has had to get past the soil storage system first. This storage system puts a significant ‘lag’ into the system, as we shall see.

The model works like this. For each year in turn, we add the calculated phosphate emissions into the storage, thus increasing the soil concentration. But we also allow a quantity of phosphate to run off to the river, the amount of run-off is not fixed, but varies in proportion to the soil concentration that year². By cycling through each year in turn, this becomes a numerical solution to the first order differential equation which I referred to in Appendix 1 of my first report.

For the next section I will focus on two specific rivers: (a) the Teme at Knighton, because it exhibits a worrying trend on EA phosphate measurements, and (b) the Ithon, because it exhibits algal blooms.

The Teme

The first step in considering time dependency is to understand how the poultry phosphate emissions has varied over time. The following graph shows the number of approved projects submitted in each year in the vicinity of Knighton:

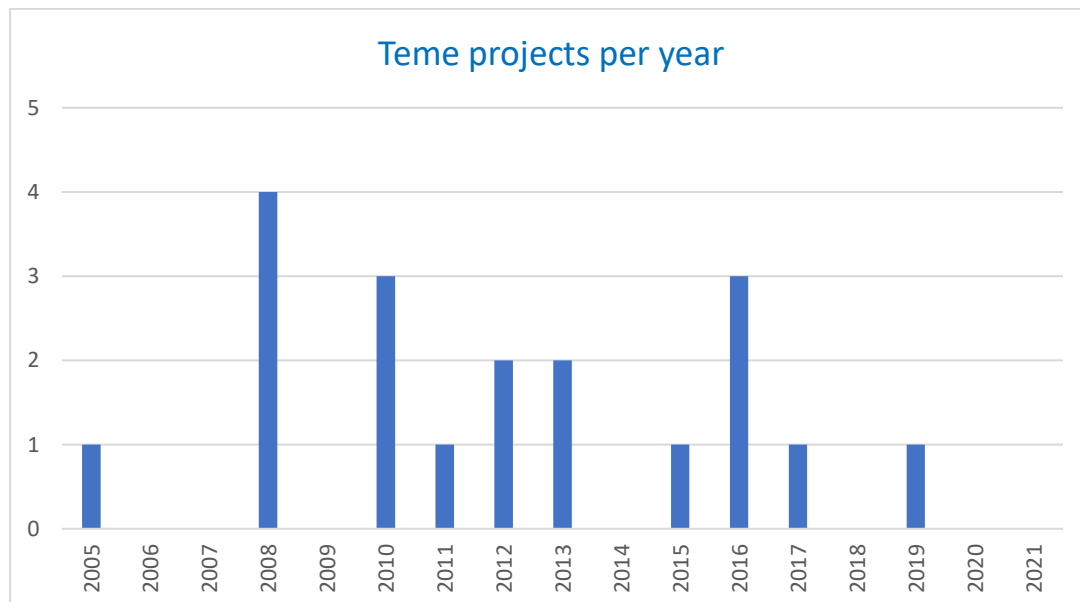


Figure 1: year of planning submission for IPU's on the river Teme

² The ratio of the capacity of the storage to the rate of run-off defines the ‘time constant’ of the system. For the following analysis, I have used a time constant of 10 years but in reality, this parameter will vary according to local conditions. 10 years will suffice for an order of magnitude calculation.

The next step is to look at the evolution of the resulting P_2O_5 emission. Although we know the date of approval, we have virtually no view of when projects come onstream. So, in the spirit of order of magnitude calculation, I have assumed a two-year lag between the year of submission and the year of commencing manure spreading. The result is the next graph:

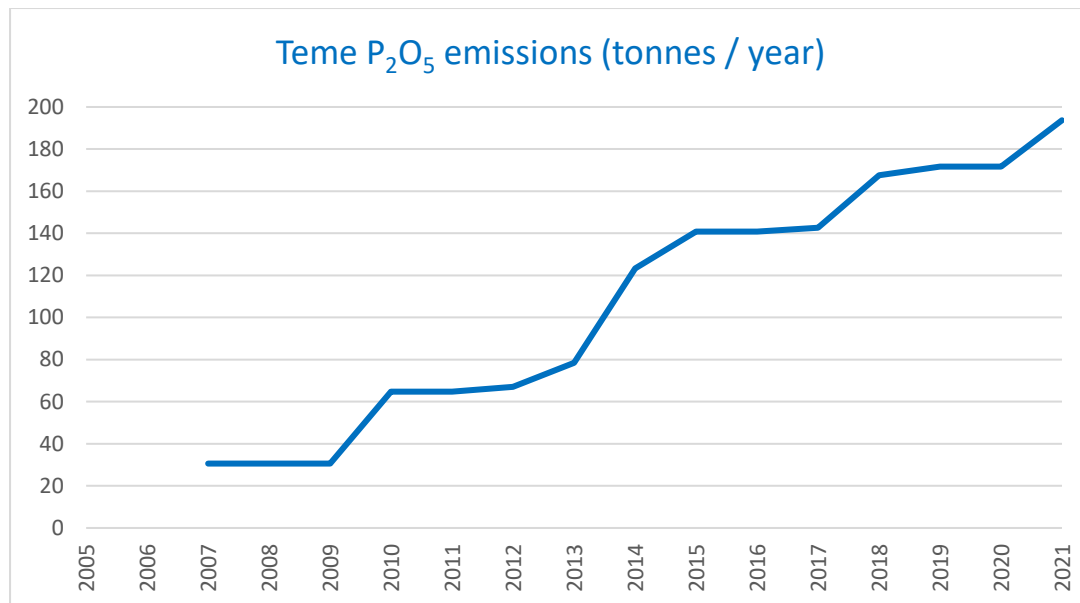


Figure 2: P_2O_5 emissions for IPUs on the river Teme

Note how the ramp-up commences in 2009 as a result of projects submitted in 2007. By 2021, P_2O_5 emissions have risen to 194 tonnes per annum. But as I pointed out above, this is not the same as the quantity of run-off to the river; this is what is being spread on the land. So now we apply the mathematical procedure described above to include the effect of soil storage and estimate the rate of run-off. The next graph is the result:

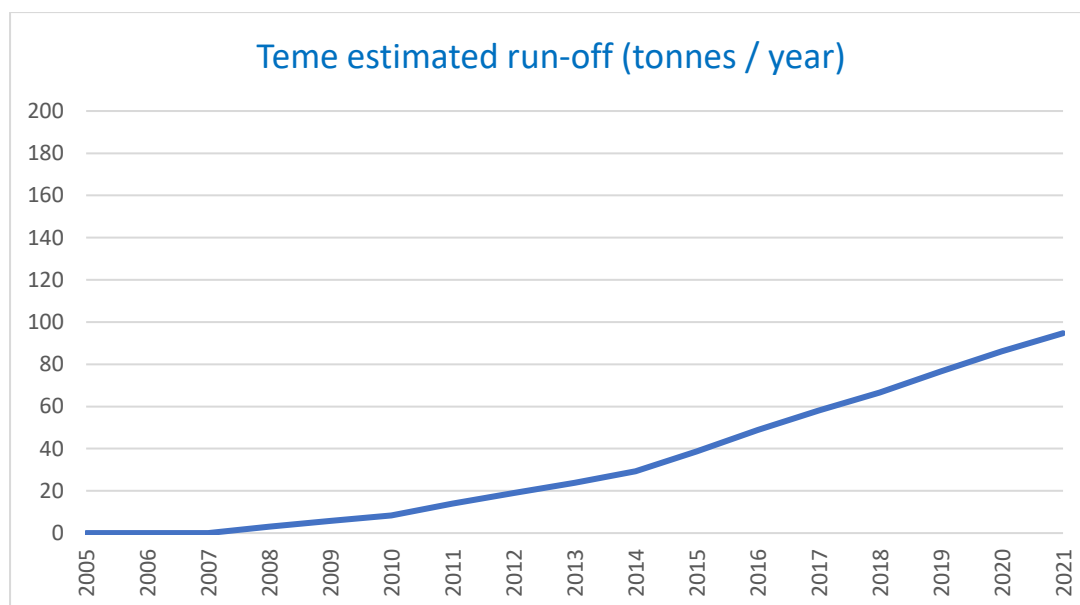


Figure 3: estimated P_2O_5 run-off due to IPUs on the river Teme

What the soil storage has done is to put a huge lag in the system. Looking again at Figure 2, we see that we reached 50% of the current emissions level around 2013-14. But 50% of peak run-off is

predicted to occur in 2021 – seven or eight years later. Make no mistake, this curve is on the way up to 194 tonnes per annum, but it will not get there for another twenty years – assuming we don't build any more IPUs.

Finally, we can translate these P2O5 run-off quantities into river P concentrations. This results in the following graph:

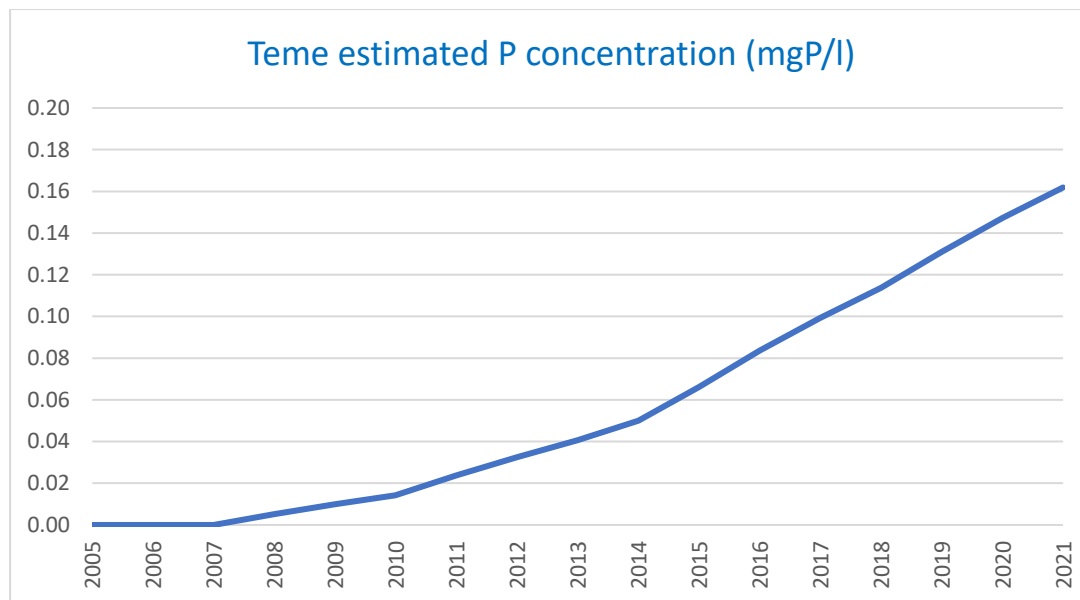


Figure 4: estimated total phosphate concentration due to IPUs on the river Teme at Knighton

Interestingly, this is a passably good fit for what the EA have reported for the Teme at Knighton: degrading from *good* to *poor* classification for phosphate over the period 2014-19.

The Ithon

Applying the same procedure for the river Ithon results in the following graphs:

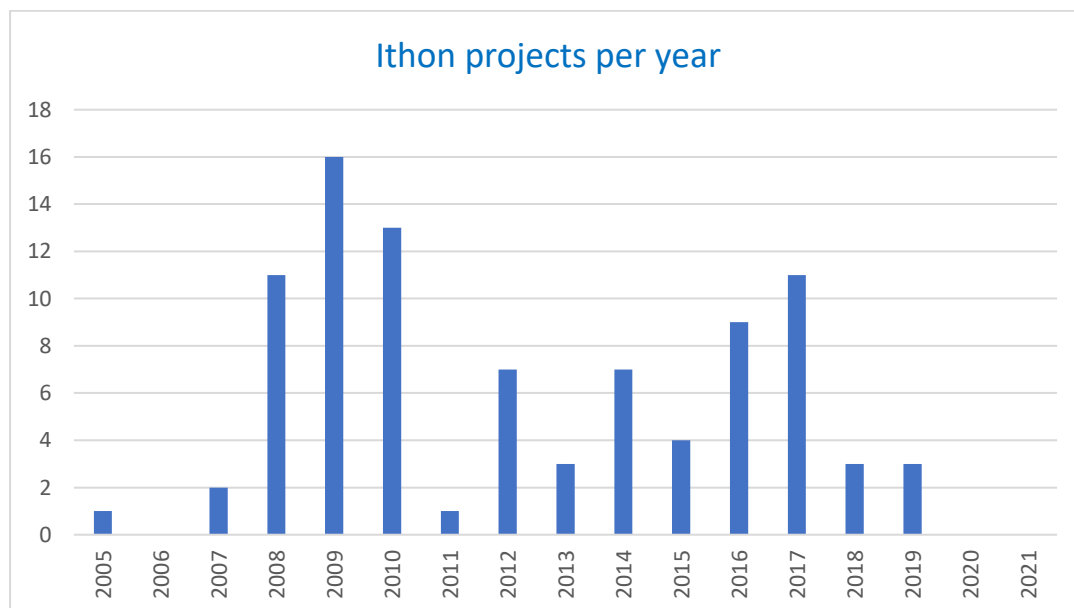


Figure 5: year of planning submission for IPUs on the river Ithon

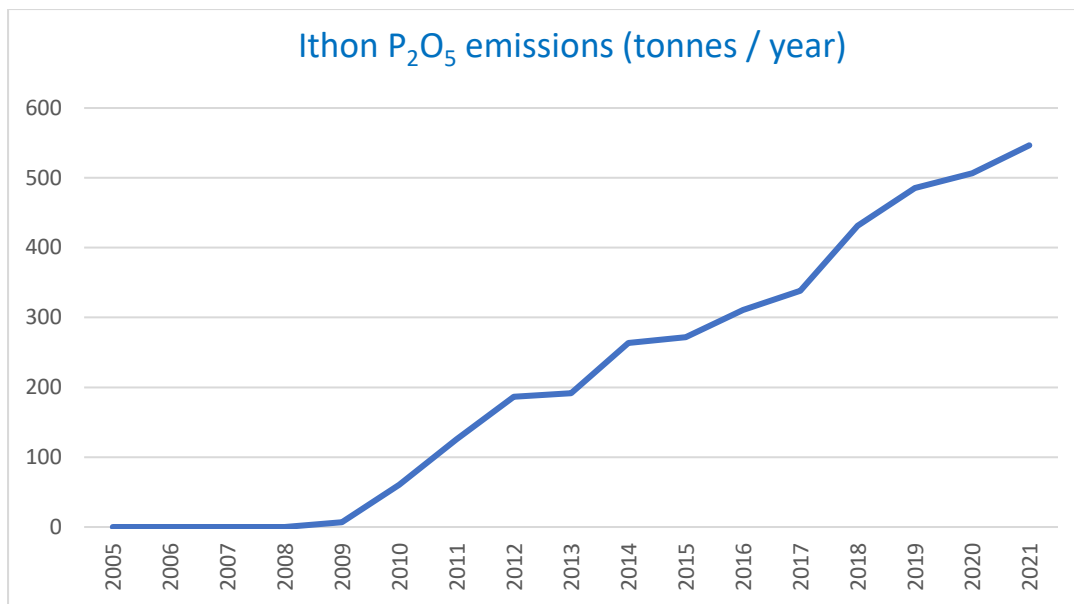


Figure 6: P_2O_5 emissions for IPU's on the river Ithon

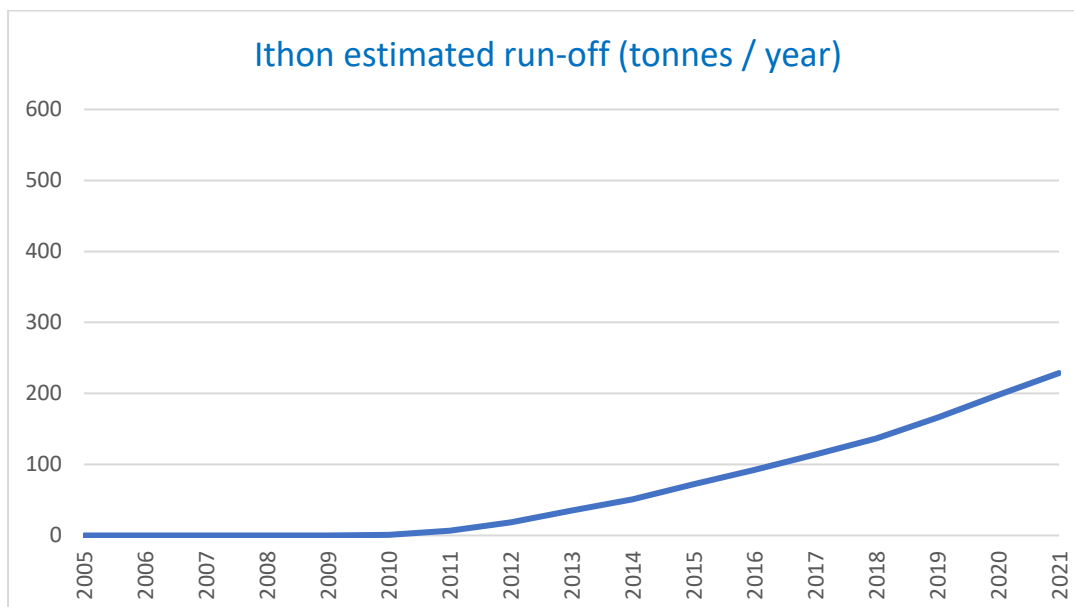


Figure 7: estimated P_2O_5 run-off due to IPU's on the river Ithon

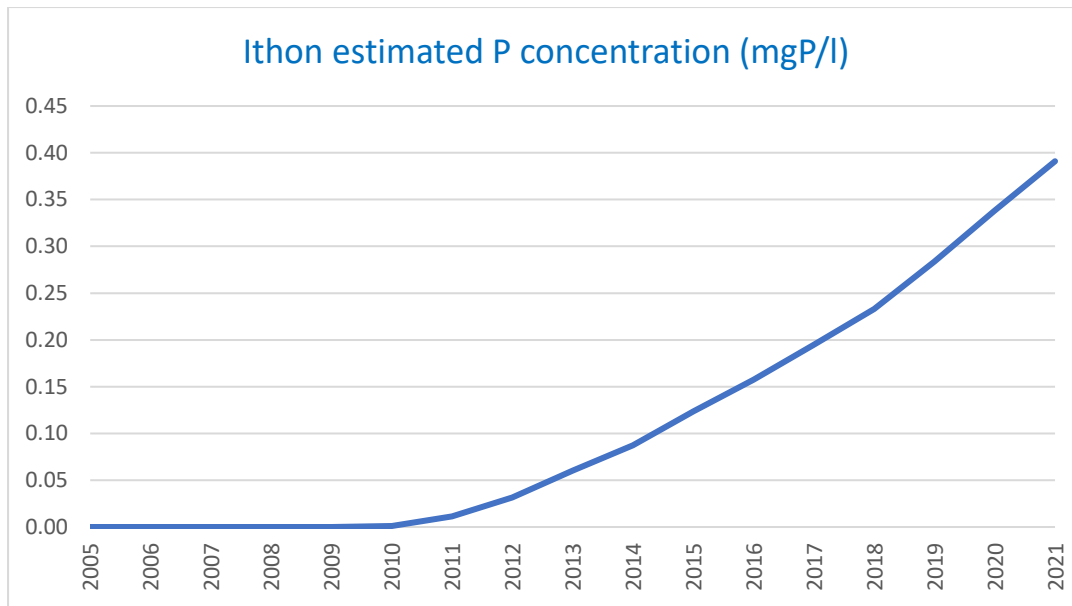


Figure 8: estimated total phosphate concentration due to IPU's on the lower river Ithon

Once again in Figure 7, we see the lag caused by soil storage. Once again in Figure 8, we see a plausible explanation for recent observations of the state of the river. In this case, the propensity for algal blooms over the last few years (as reported by WUF) is entirely consistent with the phosphate concentrations shown in Figure 8.

These two examples, the Teme and the Ithon, are typical of rivers across the two catchments in terms of their time evolution.

Summary

This has been a rather technical section, so here is a summary of the key points:

- *Equating phosphate run-off to IPU phosphate emission is overly simplistic, although it is certainly the long-term destination.*
- *We need to consider the time evolution – in particular, taking account of soil storage.*
- *It is widely known that changes to phosphate levels in soil take decades to stabilise.*
- *Soil storage introduces a huge lag between the application of phosphate to land and the resulting run-off to surface water.*
- *Current P concentrations in rivers are at maybe half the level they will achieve in another 10-15 year's time (even if we build no more IPU's).*

Conclusions

Location and nature of IPU

- In the period 2000-2019, Powys CC have approved 290 intensive poultry units to accommodate around 8.8 million birds.
- About half of IPU are free range egg producers, the rest are split between broilers, pullets and breeders. There is no significant regional segregation; all areas possess all types of IPU.

Manure disposal

- Near enough all of these IPU dispose of their poultry manure by spreading it on the land; predominantly their own or, where necessary, their neighbours' as well.
- Spreading manure as a form of waste disposal is the norm in Powys. Manure application rates are generally many times the nutrient levels required for pasture.

Planning process

- All Manure Management Plans have been approved by EA / NRW - the only requirement appearing to be compliance with the 250kgN/ha annual limit for organic fertiliser application.
- The regulators have turned a blind eye to poor nutrient management practice.
- There has been no consideration of phosphate within the planning process.

River Ithon

- The Ithon is a disaster. It carries 547 tonnes of P_2O_5 equivalent per annum, approximately 60% of the total for the Wye catchment. It is a small river, so the result is an eye-watering mean flow phosphate concentration of 0.9 mgP/litre. Here is the smoking gun with respect to the Wye algal blooms.

Time dependency of phosphate run-off

- Accumulation of phosphate in the soil is a major factor determining the timing of phosphate run-off to surface water.
- There is several year's delay between the application of phosphate and the resulting run-off.
- Today's river phosphate readings are still climbing to their equilibrium level. It is likely that today we are only half-way up the curve.
- This is still true, even if we build no more IPU.

What can we do?

- Irrespective of any change to NRW planning guidance going forwards, we have a massive legacy problem which can only be addressed by more robust regulation of the existing poultry industry.
- The recently announced reduction of the maximum application rate for organic fertiliser from 250kgN/ha to 170kgN/ha will have minimal impact on phosphate pollution.
- **There is only one solution which addresses the legacy problem:**

- *We must completely ban the practice of spreading poultry manure as a form of waste disposal and rigorously enforce best practice nutrient management.*
- *IPU operators must be forced to responsibly dispose of their hazardous waste – just like all other sectors of industry must.*

References

- [1] RB209: Nutrient Management Guide [link](#)
- [2] DEFRA guidance: July 2018,
Code of good agricultural practice (COGAP) for reducing ammonia emissions [link](#)
- [3] G. Green report, March 2021,
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